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Coursework 1: Concept Paper Development
Group F

TerraGrid Uganda

NAME	STUDENT NUMBER	REGISTRATION NO.
AGABA JOEL MUHANGUZI	2400723686	24/U/23686/PS
KASIBANTE ENOCH JAMES	2400715602	2400715602/PSA
SEMPALA HILLARY GODWIN	2400724722	24/U/24722/PS
CHELIMO EMMANUEL	2400714938	24/U/14938/PSA

PROJECT CONCEPT

The proposed project is a low cost solar powered environmental IoT platform that collects localized agricultural and soil measurements, stores them in a central database, visualizes through a web application and exposes the data through an API. A single portable sensor module will be taken to selected areas of Uganda for short data collection periods and the resulting datasets replaced through the platform to demonstrate a scalable multi-region deployment. The same module will then be installed at Makerere University for continuous live monitoring of Kampala.

INTRODUCTION

1.1 Background to the Study

Agriculture is a major part of Uganda's economy and remains highly exposed to environmental and climate variability. The World Bank reported that agriculture accounts for about 24% of Uganda's GDP and employs about 72% of the labor force. It also identifies climate shocks, soil degradation, limited irrigation and low adoption of modern inputs as important constraints to agricultural productivity.

Uganda's environmental conditions vary substantially across the country. Temperature and rainfall patterns differ by region, while droughts, floods and changing rainfall patterns create challenges for farming. The significant variation in rainfall, temperature and soil moisture can hence be classified as agricultural risks.

Soil condition is another important concern. Food and Agricultural material on the status of soil in Uganda reports that soil fertility is generally low to medium and declining with nutrient mining contributing to the problem. The same source reports relatively low fertilizer use compared with the sub saharan average. This creates a need for better localized information about soil moisture, nutrient status and environmental conditions.

The problem addressed by this project is therefore not simply the absence of sensors but the larger problem being the lack of a practical built system that can continuously collect localized environmental observations, organize them into a usable dataset, visualize changes and make the resulting data available to other applications through a standard API. The proposed platform treats the IoT device as the data acquisition layer and the web platform, database, analytics and API as the main computing contribution.

1.2 Problem Statement

Agricultural and environmental decisions making in Uganda is constrained by limited access to localized, continuously organized measurements of soil and environmental conditions. Existing information sources may provide broader or modelled datasets but a lightweight platform that collects its own localized observations, combines them with secondary datasets, provides interactive visualizations and exposes the resulting data through an API would improve the availability and reusability of environmental information. The project therefore seeks to develop and evaluate a low cost IoT to web data platform capable of collecting, storing, visualizing and analyzing environment and soil measurements in a form that can be extended to larger deployments.

1.3 Proposed Solution

The group proposes a solar powered IoT platform consisting of one portable node, a central data ingestion service, a database, a web dashboard, an API and an analytics layer. The prototype node will measure ambient temperature and Humidity, soil Temperature and moisture, soil nitrogen, phosphorus and potassium and additional soil parameters available from the selected multi parameter sensor. GPS will associate observations with locations. If multi-parameter optical probes are cost-prohibitive, the hardware uses capacitive moisture, soil temperature, and pH as primary inputs, using software models to derive soil health indicators.

2.0 PROJECT OBJECTIVES

2.1 General Objective

To design and implement a low cost solar powered IoT and web based environmental data platform for collecting, managing, visualizing and analysing localized soil and environmental measurements in Uganda

2.2 Specific Objectives

- Design and build a portable low power IoT sensor module capable of recording environmental and soil measurements
- Collect short duration field datasets from selected regions of Uganda using the same portable module.
- Develop a data ingestion and storage system that can accept both live sensor readings and replayed historical observations
- Develop a web application with a dashboard, map based visualization, historical charts and data filtering
- Develop a public facing REST API through which authorized external users or applications can retrieve project datasets.

- Implement basic analytics and machine learning experiments for spatial patterns, anomaly detection and environmental/agricultural indicators.
- Deploy the sensor module at Makerere University to provide a continuous live kampala demonstration
- Evaluate the prototype in terms of sensor operation, data reliability, platform response time, energy autonomy and usefulness of the generated visualizations.

3.0 SYSTEM OVERVIEW AND KEY FEATURES

The system will be organized into five layers: sensing, data logging/communication, backend storage, web presentation and analytics.

- IoT: ESP32 microcontroller, multi-parameter soil sensors, ambient temperature/humidity sensor and GPS.
- Solar power subsystem: small solar panel, charge controller and rechargeable battery to allow unattended operation.
- Data logging: a microsd card will store observations during field collection so that connectivity is not required while traveling.
- Data replay: field observations will be timestamped and uploaded/ replayed through the same ingestion interface used by live readings.
- Backend: a RESTful service will validate, time stamp, store and serve observations from a structured database.
- Web homepage: Current readings, latest observations, system status and navigation to the main functions
- Map page: observations displayed by geographic location with filters for parameter, data and region.
- Historical analysis page: graphs showing temperature, humidity, soil moisture and nutrient trends over time.
- API page/documentation: endpoints for retrieving observations and metadata for developers or researchers.
- Analytics: Heat maps, anomaly detection, correlation analytics and proof of concept predictive models using collected and or suitable secondary datasets.

4.0 METHODOLOGY

4.1 System development

An iterative prototyping approach will be used. The hardware will first be assembled and calibrated, followed by firmware development, backend development, database design, dashboard development and integration testing. The project will use modular components so that additional sensor nodes can be added later without redesigning the platform.

4.2 Field data collection strategy

To keep the project affordable while still demonstrating large scale applicability, the group will build one portable module rather than four permanent regional installations. The module will be taken to selected locations representing four regions of Uganda and operated for approximately two days in each location. The observations will be stored with timestamps and location information. After the field exercise, these datasets will be replayed through the platform to demonstrate how a larger network could appear in the system. The module will subsequently be installed at Makerere University for continuous live Kampala monitoring.

4.3 Data Processing and analytics

- Validate incoming measurements and flag missing, duplicated or implausible readings.
- Store raw observations together with timestamps, sensor identifiers and geographic coordinates.
- Generate descriptive statistics and time-series visualizations
- Generate spatial visualizations such as parameter heat maps from observations and suitable secondary spatial data.
- Investigate relationships between environmental variables and soil measurements.
- Develop a limited proof-of-concept machine-learning component for anomaly detection and/or prediction. Yield projection will only be attempted where sufficient historical crop/yield data can be obtained; it will not be claimed as a result of the two-day field samples alone.

4.4 Evaluation Strategy

Evaluation will cover both the physical prototype and the software platform. Hardware evaluation will consider whether measurements are consistently recorded, whether the solar battery system can support continuous operation, and whether field data can be recovered successfully. Software evaluation will consider API response time, data completeness, dashboard usability, map rendering and correctness of replayed versus live observations. Where reference instruments or credible secondary datasets are available, sensor readings will be compared against them to identify obvious measurement errors.

5.0 PROJECT JUSTIFICATION AND EXPECTED OUTCOMES

5.1 Relevance of the project

The project is relevant to Uganda because agriculture is strongly affected by climate variability, soil degradation and limited access to localized information. The design is intentionally cost-conscious. Instead of purchasing multiple permanent sensor stations, one reusable node will be used for field sampling and then converted into a permanent Kampala station. This reduces hardware cost while allowing the software architecture to demonstrate how additional nodes could be incorporated. The main contribution is therefore the complete data pipeline, from physical measurement to a reusable web and API data service rather than the number of physical sensor nodes.

5.2 Expected outcomes

- A functional solar-powered IoT sensor prototype.
- A structured database containing live and replayed environmental observations.
- A web application with homepage/navigation, dashboard, map, historical charts and analytics pages.
- A REST API for programmatic access to project data.
- A dataset containing observations from selected regions plus continuous Makerere/Kampala measurements.
- Proof-of-concept spatial and machine-learning analytics.
- Technical documentation describing the hardware, software architecture, data model and deployment process.
- Evaluation findings on the feasibility of using a low-cost portable IoT node as the basis of a larger environmental monitoring network.